

Children
lost to TB

- since you
opened this
page:

0 — almost none diagnosed

1
every
3 min ·
WHO
2024

Clark, 5, gets a chest X-ray in a mobile X-ray truck in Tondo, Manila, the
Philippines. © Ezra Acayan.

MSF TACTIC INITIATIVE · TB DIAGNOSIS IN
CHILDREN

A child dies of TB *every 3 minutes.*

Almost none of them are diagnosed.

A proven diagnostic solution exists, but it's
not being used.

THE PROBLEM · 1 OF 2

Biologically hard to detect



1

Lower bacterial loads

- Children have lower bacterial loads than adults
- Standard TB diagnostic tests designed for adults **frequently return false negatives**
- Children with TB go undiagnosed and untreated



2

Can't produce sputum

- Most diagnostics rely on sputum samples
- Most children under 5 years **can't cough up sputum on demand**
- Primary specimen collection method unfeasible for children



3

Non-specific symptoms

- Fever, fatigue, weight loss — TB symptoms in children **overlap with dozens of common illnesses**
- This causes under-diagnosis and dangerous delays



4

Extrapulmonary TB

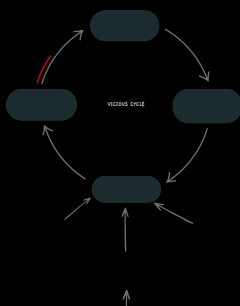
- Children get TB in **lymph nodes, bones, brain and other non-lung sites** more frequently than adults
- Sample collection and bacteriological confirmation becomes harder

neglected

Public health systems and biotechnology prioritise what is scalable, profitable and measurable, resulting in a self-reinforcing cycle that keeps children with TB sidelined.

• Identifying TB cases in hard-to-reach public health facilities
• Improving diagnosis of infectious adults

NEW DIAGNOSTIC
ALGORITHMS



A PROVEN SOLUTION

WHO Treatment Decision Algorithms:

a diagnostic flowchart that saves lives



Identifies pulmonary TB in **children under 10 years** without advanced technology or specialists



Systematises **non-specific symptoms**, TB exposure, basic test results and chest X-ray results (when available) into a cohesive scoring system



Endorsed by the **World Health Organization** in 2022



Score **> 10** triggers TB treatment — no lab confirmation required



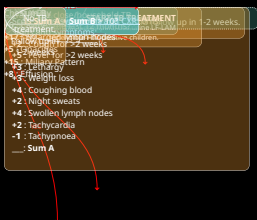
Two versions: A for clinics with chest X-ray, **B** for those without — identical logic, different scoring weights



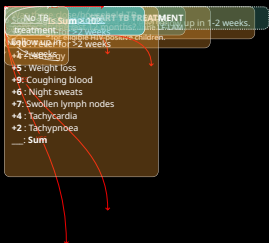
Based on data pooled from **thousands of children** across multiple countries

SWIPE THE DIAGRAM SIDWAYS TO READ IT →

Algorithm A —with chest X-ray



Algorithm B —without chest X-ray



STEP 1: ENTRY

A child under 10 years arrives at a health facility with symptoms suggesting TB. Are there **danger signs** requiring immediate stabilisation?

STEP 2: RISK ASSESSMENT

If no danger signs, the clinician assesses **risk level**. Children under 2 years, HIV-positive, or with severe acute malnutrition go straight to laboratory testing. Other children are treated for the most likely non-TB illness and followed up.

STEP 3: LAB TESTS

High-risk children and those with persistent symptoms are tested. **GeneXpert MTB/RIF** on respiratory samples. Urine **LF-LAM** for HIV-positive children. A positive result triggers treatment immediately.

STEP 4: TB DETECTED?

If the test result is available: was TB detected? **YES means treat immediately.** If negative or the test was unavailable, the algorithm continues to contact history.

STEP 5: CONTACT HISTORY

While waiting for lab test results or if they're unavailable, enquire for **close household TB contact** in the last 12 months. If yes, treat immediately. Contact history points both ways — to an undiagnosed adult source case, and to **siblings under 5 years** needing preventive treatment.

STEP 6: THE SCORING STEP · THE ONLY DIFFERENCE

If no confirmed contact: **score each symptom**. Algorithm A adds chest X-ray findings with additional points. Algorithm B (for clinics without X-ray) assigns higher weights to clinical symptoms to compensate. **Both treat if total > 10.**

STEP 7: TREATMENT

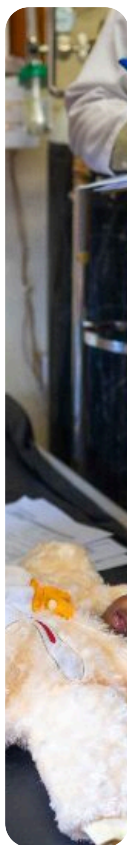
A score **above 10** triggers treatment, the same threshold for both algorithms. Children not treated for TB are reviewed in 7 to 14 days.

How do we know the WHO algorithms work? ↓

THE EVIDENCE

**1,846
children.
5 countries.
1 result.**

- MSF's TACTiC initiative conducted the **TB ALGO PED** study to investigate the diagnostic accuracy and implementability of these WHO algorithms

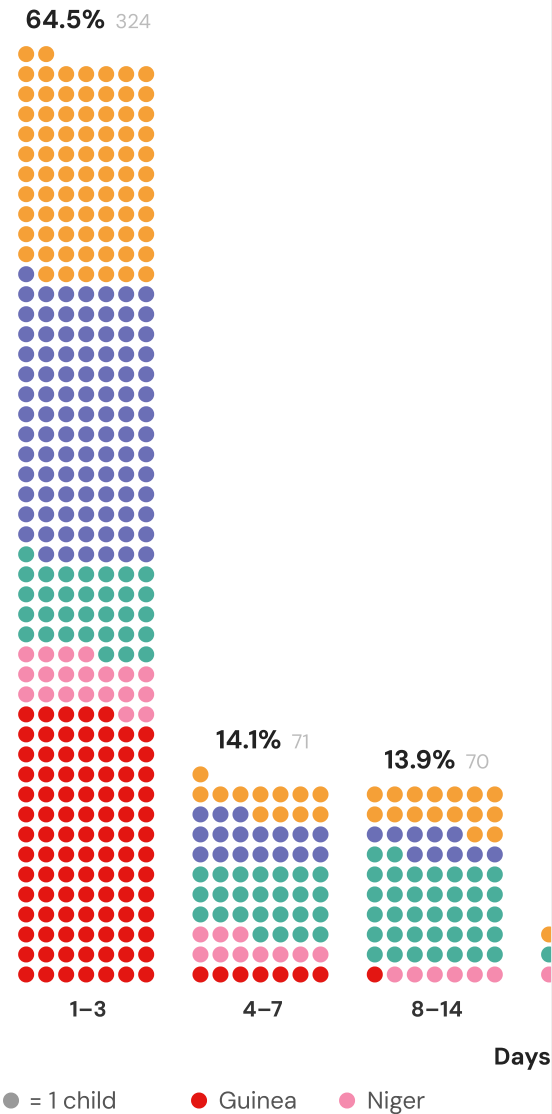


A child being evaluated for TB using the WHO algorithms in Uganda | © Stuart Tibaweswa | September 2024

- The study enrolled **1,846 children** under 10 years with symptoms consistent with TB
- Countries included:
 - Guinea
 - Niger
 - Nigeria
 - South Sudan
 - Uganda
- WHO algorithms nearly doubled the number of children initiated on lifesaving care.

Most children diagnosed within 0 days of using the WHO algorithm

Of the children who got diagnosed, consistent across all countries and all high-risk subgroups.



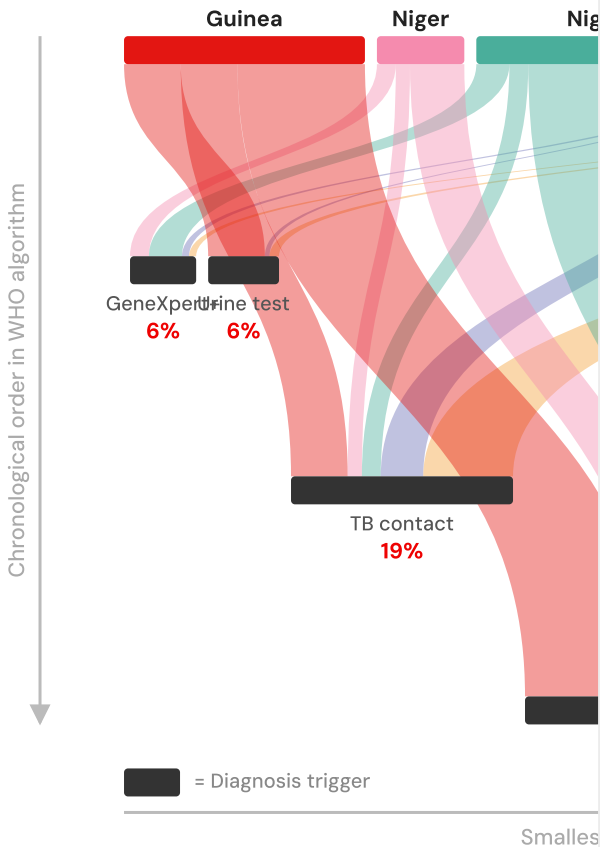
● Nigeria

● South Sudan

● Uganda

Diagnosis was mostly triggered by the ++++++ +++ in the WHO algorithms

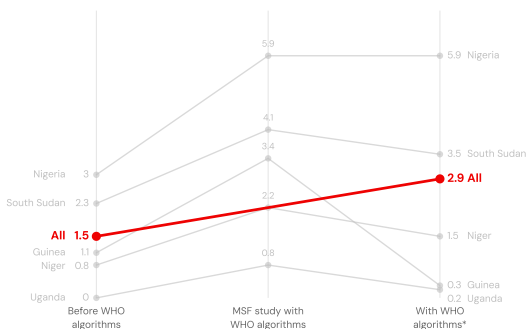
Of 496 children diagnosed using the WHO algorithm across 5 countries. Hover a flow to explore.



+ Show rare categories (clinical suspicion & other TB)

WHO algorithm use **DOUBLE** TB diagnosis rates in children

Percent (%) of children diagnosed and admitted with TB



**Uganda never implemented the WHO algorithms into their national policies*

THE FEEDBACK

Clinicians say diagnosis
with WHO algorithms is
more
*accessible, streamlined
and effective.*

“

Thanks to these algorithms, we have
diagnosed tuberculosis earlier and
saved more children in health
facilities with limited resources.



Dr Ibrahima Barry

Local Principal Investigator, Algo
TB Ped study, Guinea 🇬🇳

IMPACT

“

In an already underfunded TB response, children are pushed further to the back of the line when services are disrupted by aid cuts, conflict, or displacement.



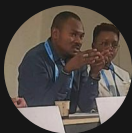
Dr Cathy Hewson

MSF Working Group Lead

IMPLEMENTATION GAP

“

The impact of implementing the WHO treatment decision algorithms in Niger has been a catalyst for hope.



**Dr Moussa Mamane
Oumarou Farouk**

TB Focal Point, Niger 🇳🇮

CALL TO ACTION

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The WHO's recommendations serve as a crucial tool to help health workers diagnose TB in children in all settings, even when there is limited access to laboratory tests and X-ray...



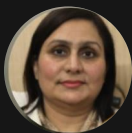
Dr Gabriella Ferlazzo

TB Medical Advisor, MSF Access Campaign

IMPACT

“

It has also significantly reduced delays in starting treatment in children. TB can spread quickly and lead to serious infections. Any delay in starting treatment can worsen their condition.



Dr Saima Umar

Head of Paediatrics, Gujranwala Teaching Hospital, Pakistan 🇵🇰

IMPACT

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The WHO algorithms have streamlined the diagnostic and treatment decision-making approach for TB in children—a process that had long remained challenging. They allow us improve the prognosis for children with TB by facilitating early diagnosis.



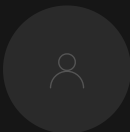
**Dr Moussa Mamane
Oumarou Farouk**

TB Focal Point, Niger 🇳🇮

DIAGNOSTIC SHIFT

“

The algorithm has opened our eyes to diagnose and treat TB in children who would have been misdiagnosed.



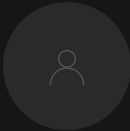
Dr Chol

Clinician, South Sudan 🇸🇸

DIAGNOSTIC SHIFT

“

Before, health workers relied on coughing... The WHO algorithms now tell us not to rely on coughing—there are other signs for TB as well.



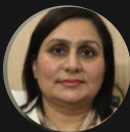
Dr Angeline Dore

TACTiC Focal Point, Guinea 🇬🇳

DIAGNOSTIC SHIFT

“

Previously, there were patients who fell into a borderline category. We suspected TB, but the tests did not show positive results... This algorithm has helped us a lot in such cases.



Dr Saima Umar

Head of Paediatrics, Gujranwala Teaching Hospital, Pakistan 🇵🇰

DIAGNOSTIC SHIFT

“

Before the WHO TB treatment decision algorithm, we used the Pakistan Paediatric Association scoring system... we often had doubts about whether we should start TB treatment.



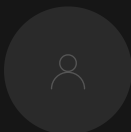
Dr Saima Umar

Head of Paediatrics, Gujranwala Teaching Hospital, Pakistan 🇵🇰

DIAGNOSTIC SHIFT

“

Using the algorithm is easy, feasible and acceptable to follow in the Malakal Project.



Dr Lazro Fidelle Nyikayo

Study Coordinator, South Sudan 🇸🇸

IMPACT

“

Our findings prove that WHO's treatment decision algorithms, for which we do not need lab test results to initiate TB treatment in children, work in real-world settings and can potentially **save many more children's lives** if implemented.



Dr Helena Huerga

Principal Investigator, TACTiC study, MSF

IMPACT

“

For algorithms to be sustained in our hospitals, **regular monitoring and follow-up is important.**



Dr Mohammed Bashir Abdullahi

Medical Activity Manager, Nigeria 

IMPLEMENTATION GAP

“

The science is clear — what's now missing is the political will to put it into practice.



Dr Helena Huerga

Principal Investigator, TACTiC study, MSF

IMPLEMENTATION GAP

“

We urge governments to adopt and put the WHO recommendations into practice... Governments must leave no stone unturned to find every child with TB and treat them right away.



Dr Gabriella Ferlazzo

TB Medical Advisor, MSF Access Campaign

CALL TO ACTION

ADDRESSING CONCERNS

Frequently Asked Questions



An MSF nurse palpates 7-year old Vedant's lymph nodes in Mumbai, India. His sister has drug-resistant TB. © Prem Hessenkamp.

How accurate are the WHO algorithms, and is there risk of over treatment?

False Positives > False Negatives — a deliberate design choice.

Sensitivity · 90.5%

Correctly identifies most children who *do* have TB — so fewer sick children are missed and go untreated.

Specificity · 83.8%

Some children who *don't* have TB will still be flagged — a deliberate tradeoff. **Missing a child in a disease this deadly costs more than over-treating.**

Aren't lab tests more reliable than the WHO algorithms?

Not in children. Current lab tests often fail to detect pediatric TB. WHO algorithms use clinical, radiographic, and contact-with-TB evidence to reflect how diagnosis actually happens in practice.

Are health systems equipped to use WHO algorithms?

Yes. WHO algorithms are designed for low-resource, primary care settings and reduce reliance on laboratory infrastructure.

Is there enough evidence to support WHO algorithms?

Yes. Endorsed by the World Health Organization in 2022, WHO algorithms are backed by multi-country evidence showing strong diagnostic performance and real-world impact.

How complex is implementation?

No special devices or new equipment are needed. Rolling them out requires staff training, WHO algorithm integration into existing workflows and anticipating drug needs for treatment.

However, the main barrier is policy inertia and incentive structures that favor lab-confirmed cases over clinically diagnosed children.

ACT NOW

**The WHO algorithms are
ready.
We know they work.
*Children are waiting.***



Vaishnavi, 7, has DR-TB and interacts with MSF nurse Prachi in Mumbai, India. February 2022. © Prem Hessenkamp.

The WHO algorithms are validated in the field and
can be integrated into National TB Programmes
today.

While you read this, approximately **0** more children
died of TB — almost none diagnosed. A preventable,

treatable disease with a proven diagnostic solution.



Data: MSF TACTiC operational study (TB-ALGO-PED),

Epicentre

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